Qatar Labor Laws RAG Chatbot

Intelligent AI-Powered Legal Assistant MVP

Executive Summary

An advanced RAG (Retrieval-Augmented Generation) chatbot that provides accurate, instant answers to Qatar Labor Law queries. Built with enterprise-grade architecture combining MCP servers, vector databases, and modern AI frameworks.

System Architecture

The MVP implements a sophisticated multi-tier architecture designed for accuracy, scalability, and user experience:

Layer	Component	Technology
1. User Interface	Web-based Chat Interface	Gradio 4.x
2. Communication	MCP Server Protocol	n8n MCP + SSE Streaming
3. Orchestration	Workflow Engine	n8n Subworkflow
4. AI Processing	RAG Pipeline	Vector Search + LLM
5. Data Layer	Vector Database	Supabase (PostgreSQL + pgvector)

Technical Data Flow

Step	Process
1	User submits question via Gradio web interface
2	Frontend sends query to n8n MCP server using SSE protocol
3	MCP server triggers RAG subworkflow in n8n
4	Query is converted to vector embeddings
5	Supabase vector search retrieves relevant Qatar Labor Law sections
6	Retrieved context + query sent to LLM for answer generation
7	LLM generates accurate, context-aware response
8	Response flows back: n8n → MCP → Gradio → User

Core Technologies

MCP (Model Context Protocol): Anthropic's protocol for AI-server communication with real-time streaming

RAG Architecture: Retrieval-Augmented Generation ensures factual, grounded responses

Supabase pgvector: High-performance vector similarity search for semantic matching

n8n Workflow: Visual automation platform orchestrating the entire AI pipeline

Key Features & Capabilities

Feature	Description	Benefit
Semantic Search	Vector-based query matching	Finds relevant laws even with different wording
Context-Aware AI	LLM with retrieved legal context	Accurate, legally-grounded answers
Real-time Responses	SSE streaming protocol	Instant query results under 3 seconds
Scalable Architecture	Microservices-based design	Handles multiple concurrent users
User-Friendly UI	Modern chat interface	No technical knowledge required

Real-World Use Cases

- **HR Departments:** Quickly verify leave policies, working hours, and termination procedures
- **Legal Teams:** Research specific labor law clauses and employee rights
- **Employees:** Self-service access to understand their rights and entitlements
- **Recruiters:** Accurate information on probation periods and contract requirements

Technical Advantages

Aspect	Implementation
Accuracy	RAG ensures responses are grounded in actual Qatar Labor Law documents
Performance	Vector search optimized for sub-second retrieval from thousands of law sections
Maintainability	Modular n8n workflows enable easy updates without code changes
Extensibility	MCP architecture allows adding new data sources or AI capabilities seamlessly
Deployment	Cloud-ready on Hugging Face Spaces with automatic scaling

Business Value Proposition

This MVP demonstrates enterprise-ready AI capabilities with immediate ROI through automated legal query resolution. The system reduces dependency on legal consultations for routine questions, provides 24/7 availability, and ensures consistency in labor law interpretation across the organization. The modular architecture enables rapid expansion to other legal domains or regulatory frameworks.

MVP Status: Production-Ready | **Deployment:** Cloud-based (Hugging Face Spaces) | **Integration:** API-ready for enterprise systems